1 RelCalc report for spin system: NH

Spins: N_1:$+\frac{1}{2}$ H_2:$+\frac{1}{2}$
Interactions:

Dipolar between H_2 and N_1 $\beta = \beta$ $\alpha = 0$ distance R_d constant $d = \frac{\mu_0 \hbar \gamma_H \gamma_N}{4\pi (R_d)^3}$, local motion: static

CSA for N_1 $\beta =$ $\alpha = 0$ constant $c_A = \gamma_N B_0 \Delta\delta_{sym}/3$, $\Delta\delta_{sym} = \delta_{zz} - \delta_{yy}$, local motion: static

CSA for H_2 $\beta = 0$ $\alpha = 0$ constant $c_X = \gamma_H B_0 \Delta\delta_{sym}/3$, $\Delta\delta_{sym} = \delta_{zz} - \delta_{yy}$, local motion: static

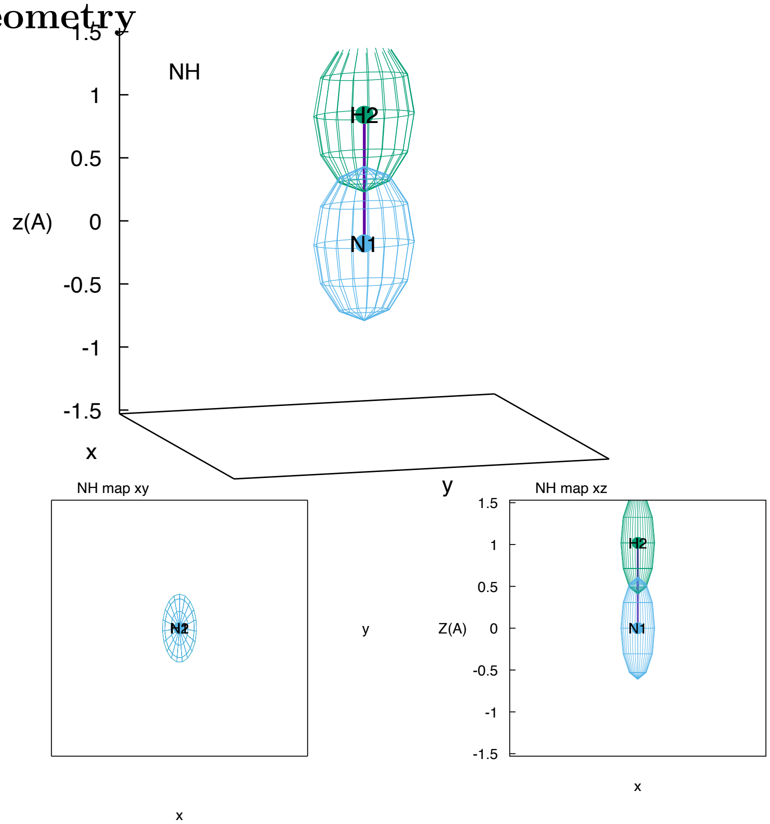
Symbolic factors:
Legendre polynomials: β: (rank 2; P_q) : (rank 2; Q_q).
Macromolecular terms only: False
Spectral density functions: Isotropic global tumbling (ISO):

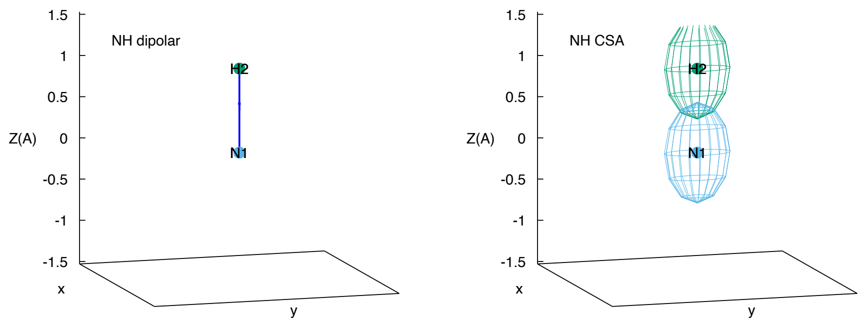
$$J(q, \omega) = \frac{\tau(q)}{1 + \omega^2 \tau(q)^2}$$

Static timescale:

$$\tau(0) = \tau_c$$

2 Geometry





The geometry of the NH spin system evalulated for current parameters. The AX dipolar interaction from the first spin is used to calculate the geometry. Arrows on the dipolar interactions should indicate direction. Otherwise indicates an error (other than 'ext' interactions). Axially symmetric tensoral components are represented by an ellipse. Units are Angstroms.

3 Relaxation rates

Rate for $N1p$ on $N1p$:

$$R_{N1p,N1p} = \begin{aligned} &+c_A^2\left[+\tfrac{4}{5}\tau_c + \tfrac{3}{5}J\left(0,\omega_N\right)\right] \\ &+d^2\left[+\tfrac{1}{5}\tau_c + \tfrac{1}{20}J\left(0,\omega_H-\omega_N\right) + \tfrac{3}{20}J\left(0,\omega_N\right) + \tfrac{3}{10}J\left(0,\omega_H\right) + \tfrac{3}{10}J\left(0,\omega_H+\omega_N\right)\right] \end{aligned} \tag{1}$$

Rate for $N1z$ on $N1z$:

$$R_{N1z,N1z} = \begin{aligned} &+c_A^2\left[+\tfrac{6}{5}J\left(0,\omega_N\right)\right] \\ &+d^2\left[+\tfrac{1}{10}J\left(0,\omega_H-\omega_N\right) + \tfrac{3}{10}J\left(0,\omega_N\right) + \tfrac{3}{5}J\left(0,\omega_H+\omega_N\right)\right] \end{aligned} \tag{2}$$

Rate for $H2z$ on $N1z$:

$$R_{N1z,H2z} = +d^2\left[-\tfrac{1}{10}J\left(0,\omega_H-\omega_N\right) + \tfrac{3}{5}J\left(0,\omega_H+\omega_N\right)\right] \tag{3}$$

Rate for $N1z$ on $H2z$:

$$R_{H2z,N1z} = +d^2\left[-\tfrac{1}{10}J\left(0,\omega_H-\omega_N\right) + \tfrac{3}{5}J\left(0,\omega_H+\omega_N\right)\right] \tag{4}$$

Rate for $H2z$ on $H2z$:

$$R_{H2z,H2z} = \begin{aligned} &+c_X^2\left[+\tfrac{6}{5}J\left(0,\omega_H\right)\right] \\ &+d^2\left[+\tfrac{1}{10}J\left(0,\omega_H-\omega_N\right) + \tfrac{3}{10}J\left(0,\omega_H\right) + \tfrac{3}{5}J\left(0,\omega_H+\omega_N\right)\right] \end{aligned} \tag{5}$$

Rate calculation time: 0:00:00.000657

4 Parameter index

Key	Value	Description
d:	(1.02e-10, 'N', 'H')	dipolar between H_2 and N_1(distance+nuclei)
beta:	0.00^o	angle for Legendre polynomials: P
betaC:	0.00^o	angle for Legendre polynomials: Q
cX:	(10.0, 'H')	CSA for H_2 (ppm+nucleus)
cA:	(-160.0, 'N')	CSA for N_1 (ppm+nucleus)

5 Plots at specified values

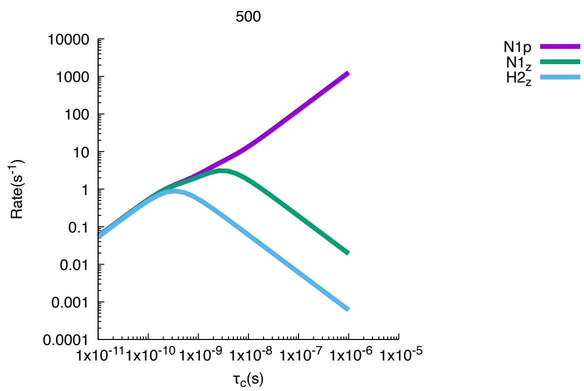


Figure 1: Relaxation rates versus correlation time τ_c for spin system NH. macro=False. ν_0 500.00 MHz. β=0.00^o (P rank 2). =0.00^o (Q rank 2). Dipolar d: (γ_N, γ_H), R_d=1.02Å, d=7.21e+04 rad s^{-2}. CSA: c_A (γ_N) -160.00ppm, c_A=1.70e+04 rad s^{-2}. CSA: c_X (γ_H) 10.00ppm, c_X=1.05e+04 rad s^{-2}.